

Decision Tree classifier

A **decision tree** is a **supervised machine learning algorithm** used for **classification** and **regression**.

It works by **splitting** data into **branches** based on **feature conditions** to arrive at a decision.

Example

Sunny?	Temperature > 20°C?	Play Outside?
Yes	Yes	Yes
Yes	No	No
No	-	No

Imagine you are deciding whether to play outside. You ask two questions:

Is it sunny?

If **yes**, go to the next question.

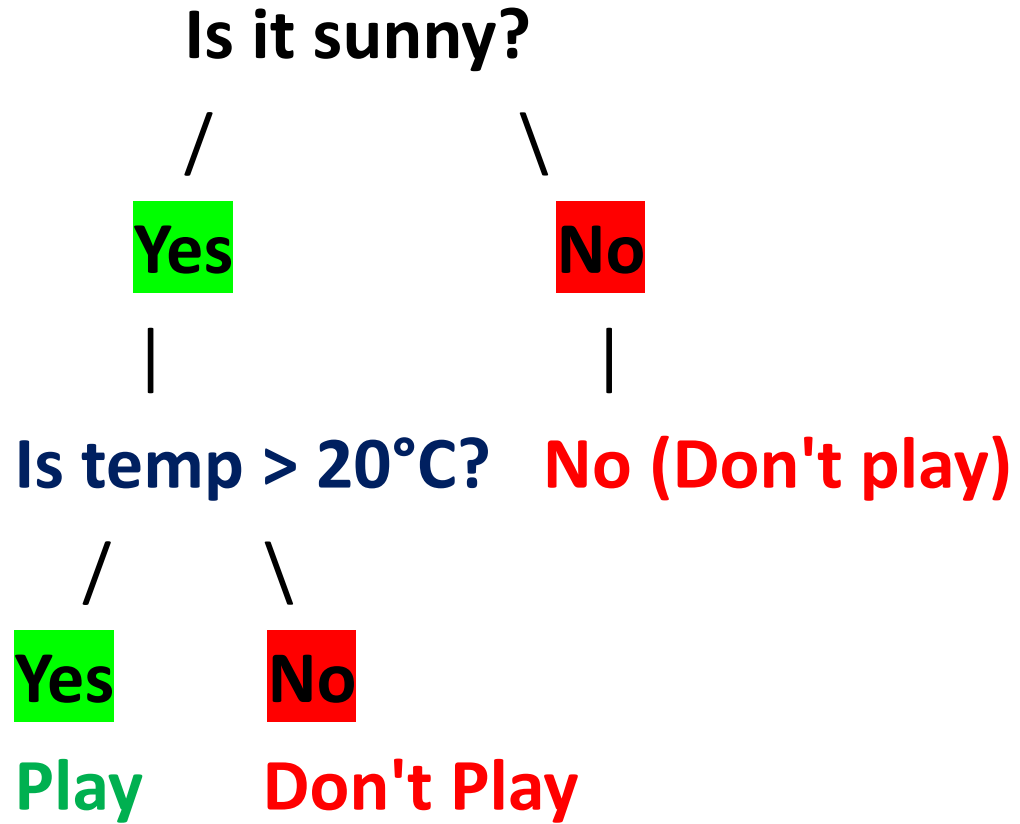
If **no**, the answer is **No** (you won't play outside).

Is the temperature above 20°C?

If **yes**, the answer is **Yes** (you will play outside).

If **no**, the answer is **No**.

Decision Tree



Algorithms to build decision tree

- CART (**Classification and Regression Trees**)
- ID3 (**Iterative Dichotomiser 3**) are two popular algorithms used for building decision trees

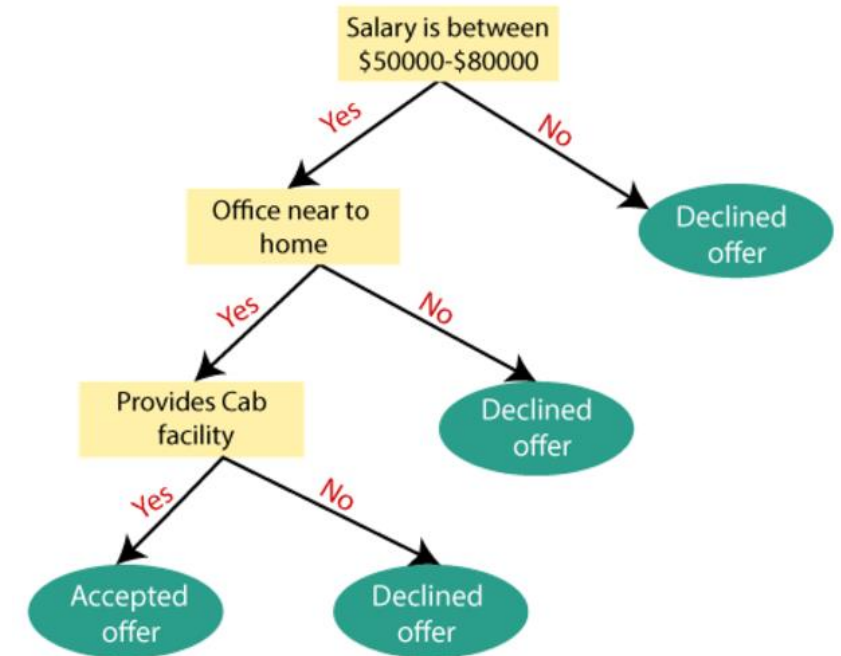
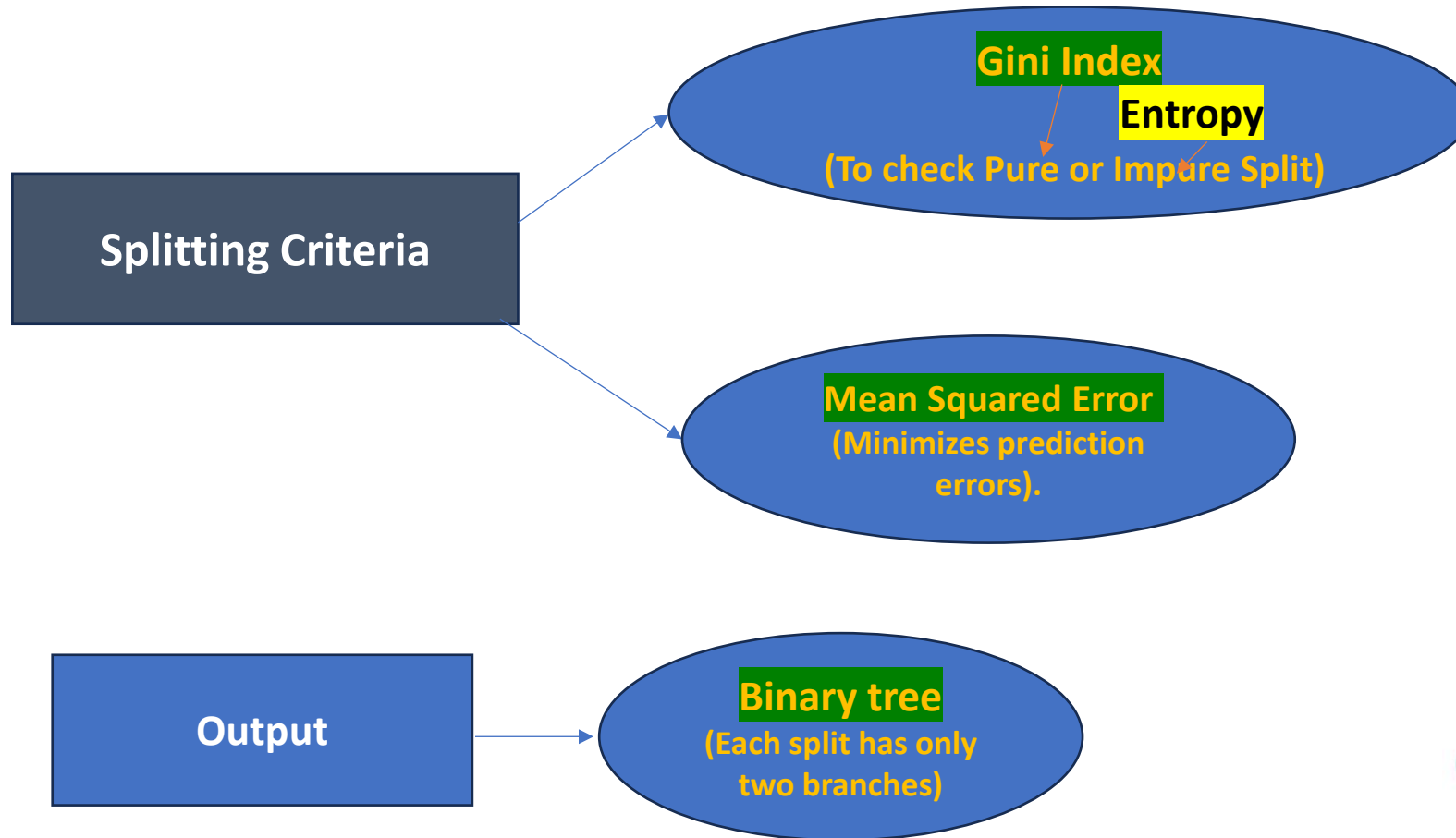
Example Use Cases:

CART: Predicting house prices (**regression**) or spam email detection (**classification**).

ID3: Deciding if a customer will buy a product based on **categorical features** like "**Job: [Yes/No]**" or "**Marital Status.**"

CART (Classification and Regression Trees):

Builds a Decision Tree for both **Classification and Regression**



Works well for continuous data

ID3 (Classification tree):

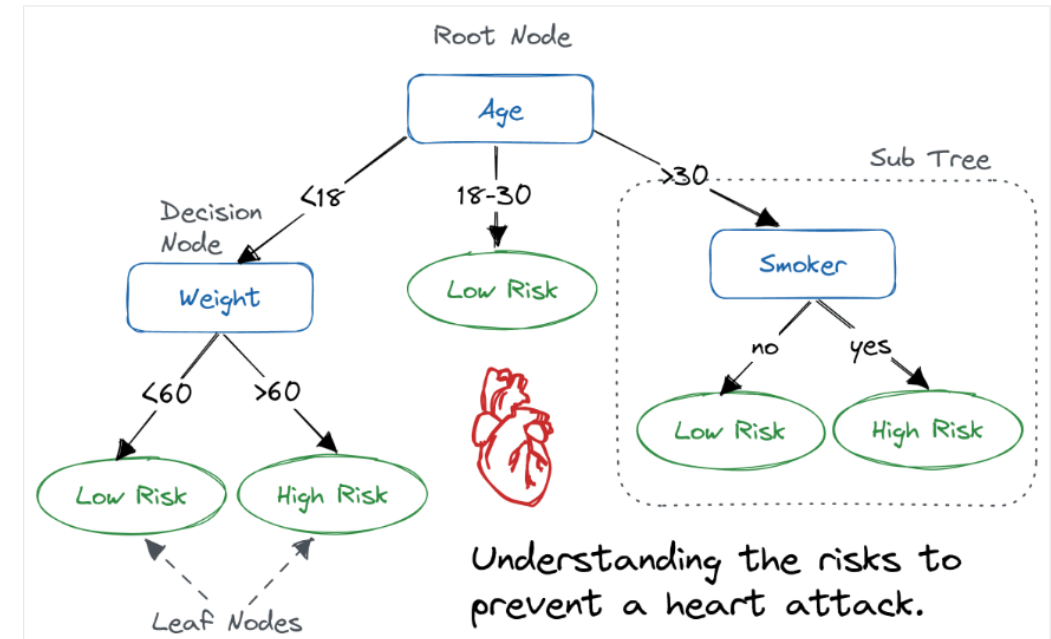
Builds a Decision Tree for **Classification only**

Splitting Criteria

Information Gain
To select best
feature

Output

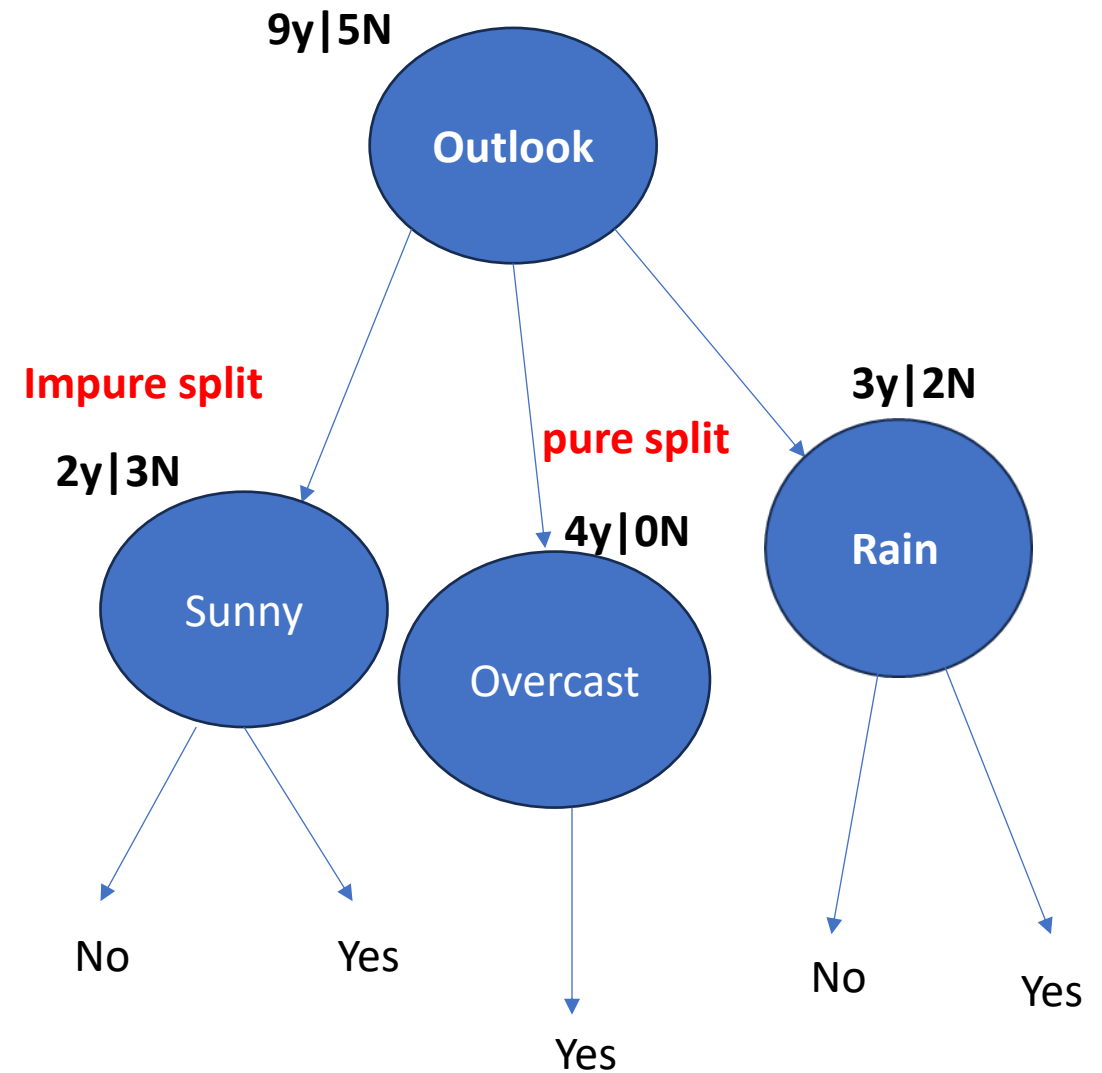
Multiway tree
(can have more than
two branches at each
split)



Works well for categorical data

Pure Impure split Example

No	Outlook	Temperature	Humidity	Windy	Play
1	Sunny	Hot	High	FALSE	No
2	Sunny	Hot	High	TRUE	No
3	Overcast	Hot	High	FALSE	Yes
4	Rain	Mild	High	FALSE	Yes
5	Rain	Cool	Normal	FALSE	Yes
6	Rain	Cool	Normal	TRUE	No
7	Overcast	Cool	Normal	TRUE	Yes
8	Sunny	Mild	High	FALSE	No
9	Sunny	Cool	Normal	FALSE	Yes
10	Rain	Mild	Normal	FALSE	Yes
11	Sunny	Mild	Normal	TRUE	Yes
12	Overcast	Mild	High	TRUE	Yes
13	Overcast	Hot	Normal	FALSE	Yes
14	Rain	Mild	High	TRUE	No



An **Pure Split** occurs when all data points in a branch belong to a single class (e.g., all "Yes" or all "No").

An **Impure Split** occurs when a branch contains a mix of classes.

Entropy

$$\begin{aligned}\text{Entropy} &= - \sum_{i=1}^k p_i \log_2 p_i \\ &= -(p_+ \log_2 p_+) - (p_- \log_2 p_-)\end{aligned}$$

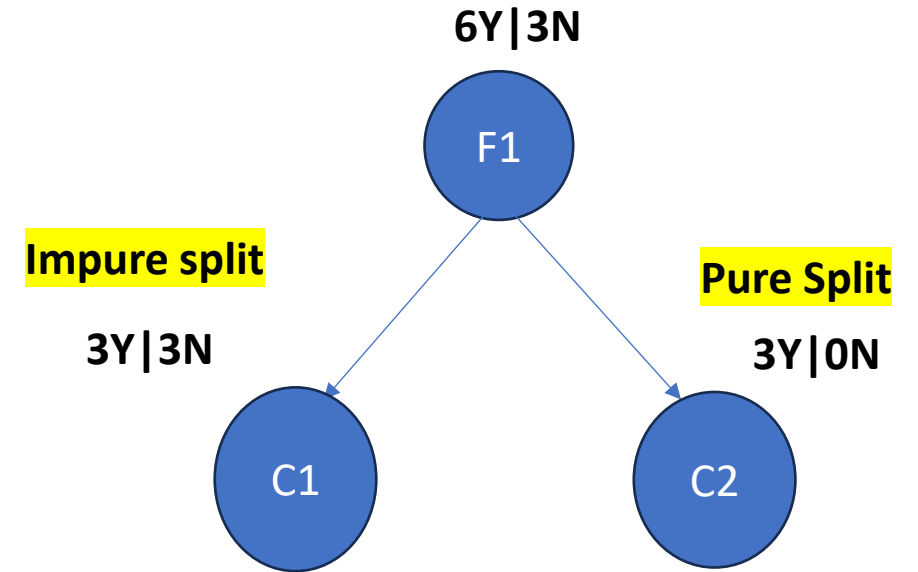
Where:

k : Number of classes.

p_i : Proportion of instances in class i .

Example:

- If a dataset has 2 classes, "Yes" (60%) and "No" (40%)
 p_+ p_-

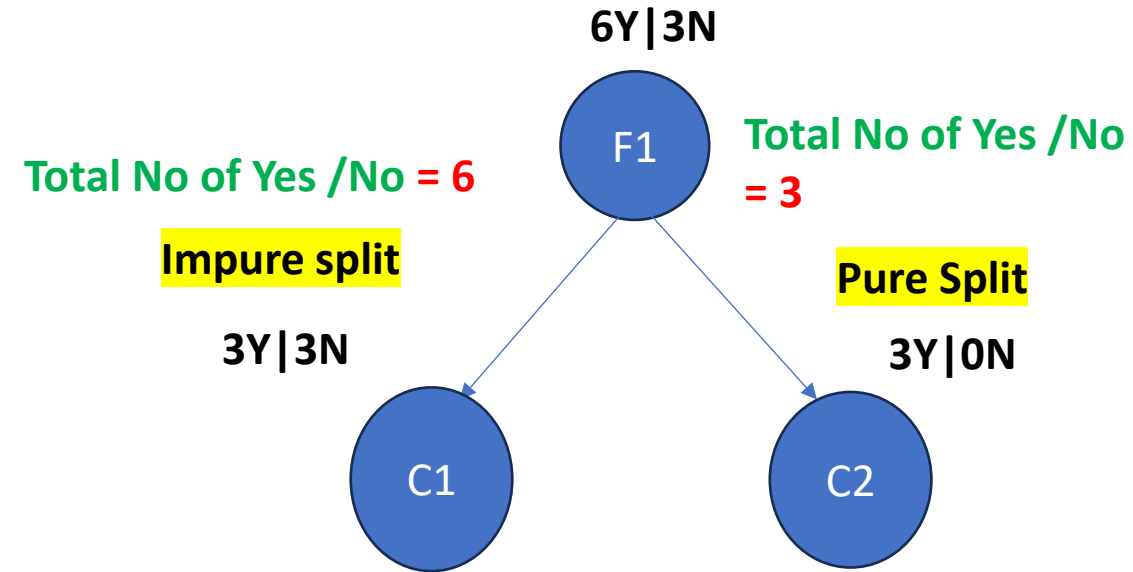


Entropy

$$\begin{aligned}\text{Entropy} &= - \sum_{i=1}^k p_i \log_2 p_i \\ &= -(p_+ \log_2 p_+) - (p_- \log_2 p_-)\end{aligned}$$

$$\begin{aligned}\text{Entropy}(c1) &= -(p_+ \log_2 p_+) - (p_- \log_2 p_-) \\ &= -\left(\frac{3}{6} \log_2 \frac{3}{6}\right) - \left(\frac{3}{6} \log_2 \frac{3}{6}\right) \\ &= 1 \quad \text{this is the maximum impure split value}\end{aligned}$$

$$\begin{aligned}\text{Entropy}(c2) &= -(p_+ \log_2 p_+) - (p_- \log_2 p_-) \\ &= -\left(\frac{3}{3} \log_2 \frac{3}{3}\right) - \left(\frac{0}{3} \log_2 \frac{0}{3}\right) \\ &= 0 \quad \text{this is pure split}\end{aligned}$$



.....

For binary classification ($k=2$), **Entropy** ranges from **0 to 1**.

For more than 2 classes ($k>2$), **Entropy** range increases to $[0, \log_2(k)]$

As k increases, $\log_2(k)$ grows, so the range of entropy also increases.

For $k= 2$: $\log_2(2) = 1$

For $k= 3$: $\log_2(3) = 1.585$

For $k= 4$: $\log_2(4) = 2$

For $k= 8$: $\log_2(8) = 3$

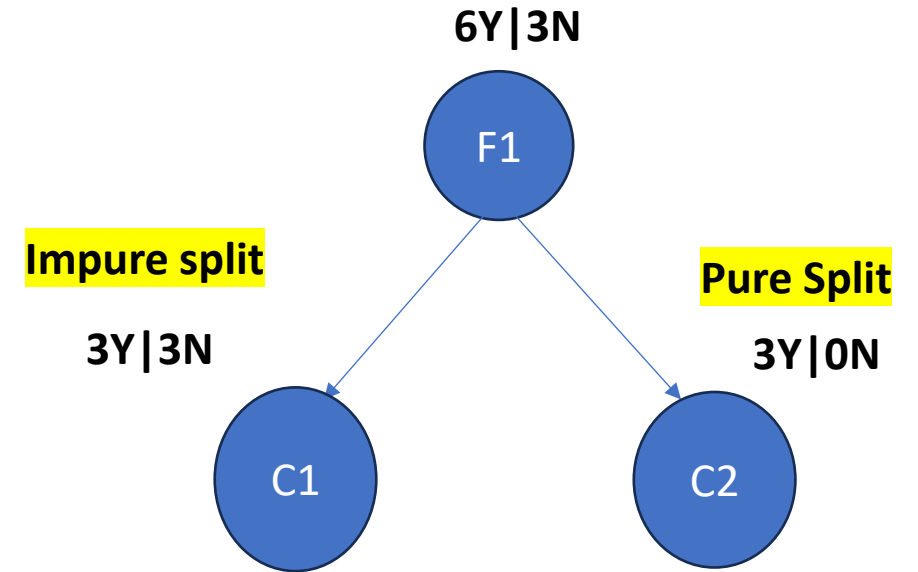
Gini Impurity or Gini Index

$$\text{Gini} = 1 - \sum_{i=1}^k p_i^2$$
$$= 1 - ((p_+)^2 + (p_-)^2)$$

$$\begin{aligned}\text{Gini}(c1) &= 1 - \left(\left(\frac{3}{6}\right)^2 + \left(\frac{3}{6}\right)^2\right) \\ &= 1 - \left(\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right) \\ &= 0.5 \text{ (complete impure split)}\end{aligned}$$

$$\begin{aligned}\text{Gini}(c2) &= 1 - \left(\left(\frac{3}{3}\right)^2 + \left(\frac{0}{3}\right)^2\right) \\ &= 1 - 1 \\ &= 0 \text{ (pure split)}\end{aligned}$$

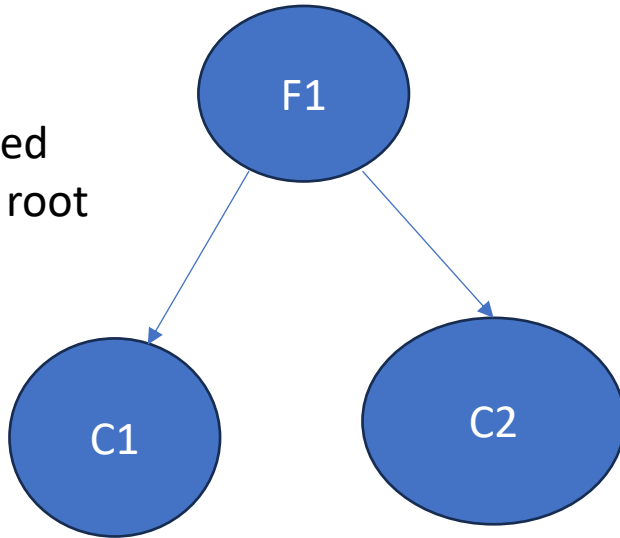
For binary classification ($k=2$), Gini ranges from 0 to 0.5.



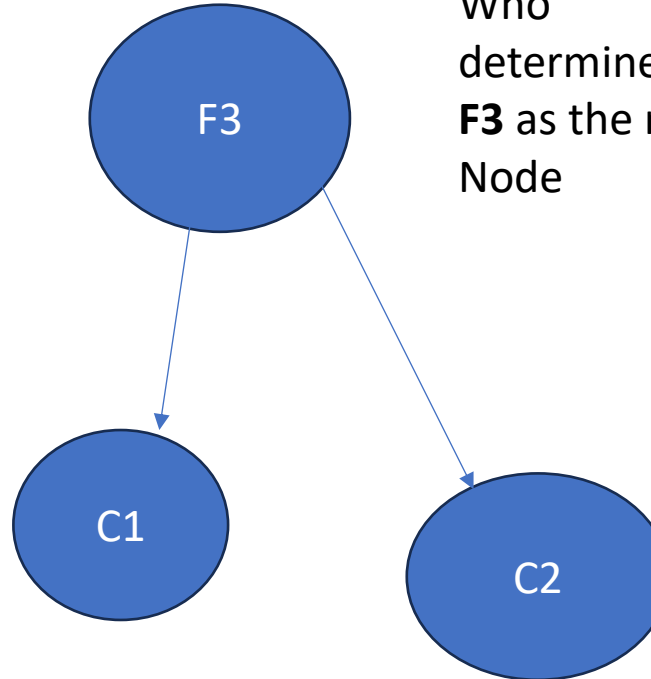
Information Gain

Suppose I have three features **f1 ,f2,f3**

Who determined **F1** as the root Node.



Who determined **F3** as the root Node



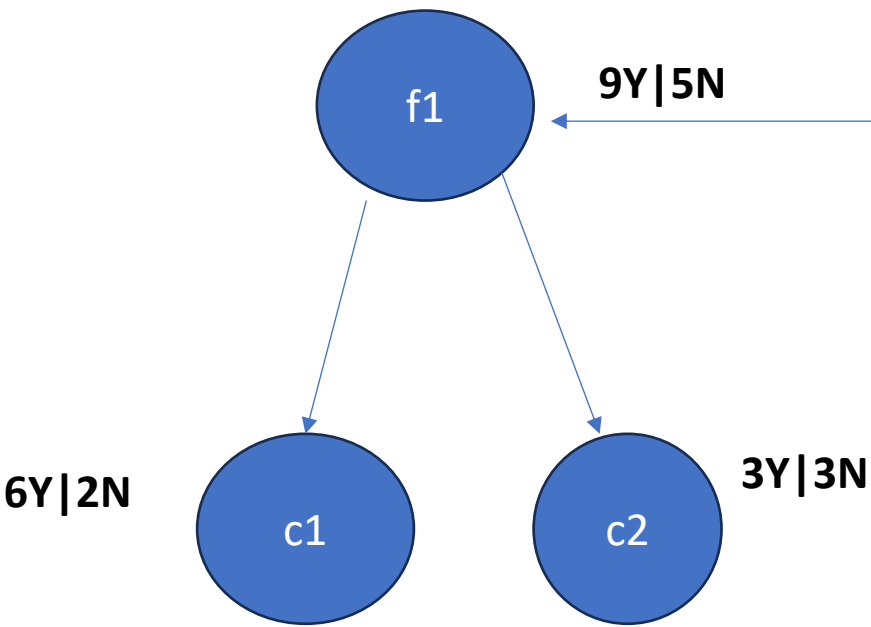
Information Gain

- It **identifies** the feature that provides the **maximum clarity** about the **target variable**.
- Efficiently **splitting the features** to create **most effective decision tree**.

Information Gain:

$$\text{Information Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \left(\frac{|S_v|}{|S|} \cdot \text{Entropy}(S_v) \right)$$

Humidity
High
High
High
High
Normal
Normal
Normal
High
Normal
Normal
Normal
High
Normal
High

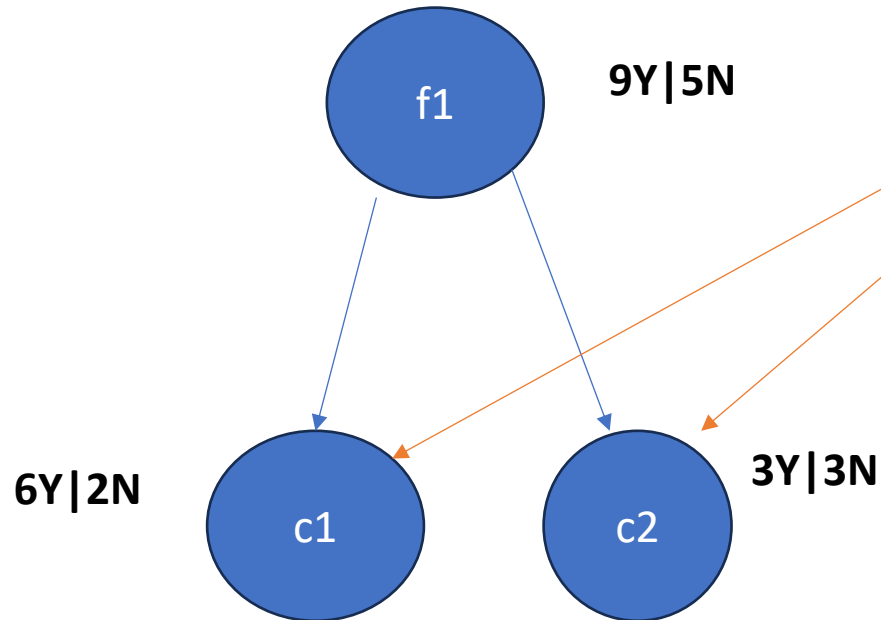


Entropy(s) = $-(p_+ \log_2 p_+) - (p_- \log_2 p_-)$
= $-(\frac{9}{14} + \log_2 \frac{9}{14}) - (\frac{5}{14} + \log_2 \frac{5}{14})$
= 0.94

Information Gain:

$$\text{Information Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \left(\frac{|S_v|}{|S|} \cdot \text{Entropy}(S_v) \right)$$

Entropy of features C1 and C2

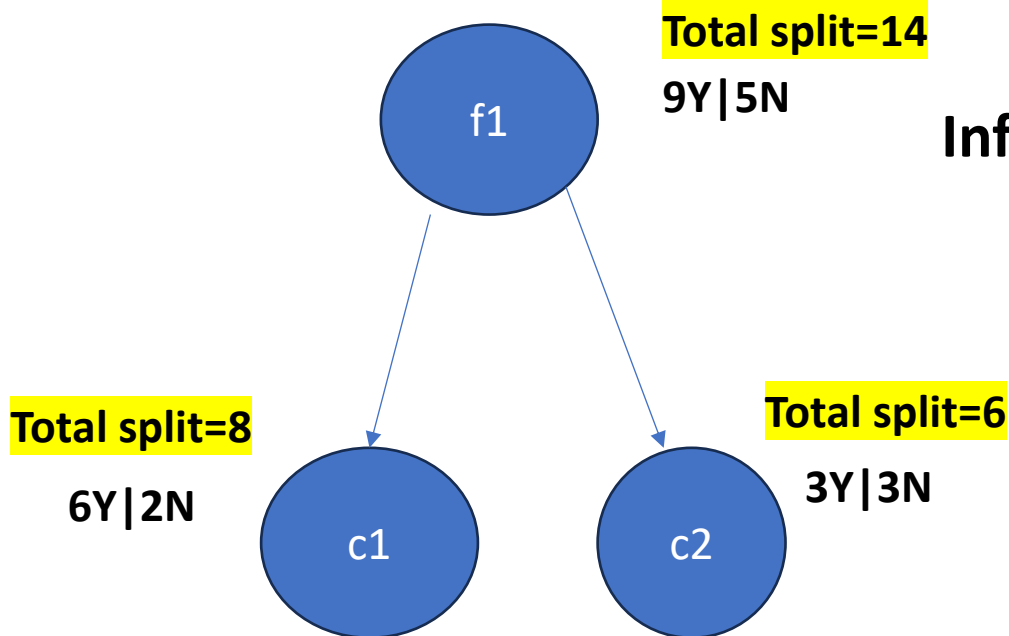


$$\begin{aligned} \text{Entropy}(c_1) &= -(p_+ \log_2 p_+) - (p_- \log_2 p_-) \\ &= -\left(\frac{6}{8} + \log_2 \frac{6}{8}\right) - \left(\frac{2}{8} + \log_2 \frac{2}{8}\right) \\ &= \mathbf{0.81} \end{aligned}$$

$$\begin{aligned} \text{Entropy}(c_2) &= -(p_+ \log_2 p_+) - (p_- \log_2 p_-) \\ &= -\left(\frac{3}{6} + \log_2 \frac{3}{6}\right) - \left(\frac{3}{6} + \log_2 \frac{3}{6}\right) \\ &= \mathbf{1} \end{aligned}$$

Information Gain:

$$\text{Information Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \left(\frac{|S_v|}{|S|} \cdot \text{Entropy}(S_v) \right)$$

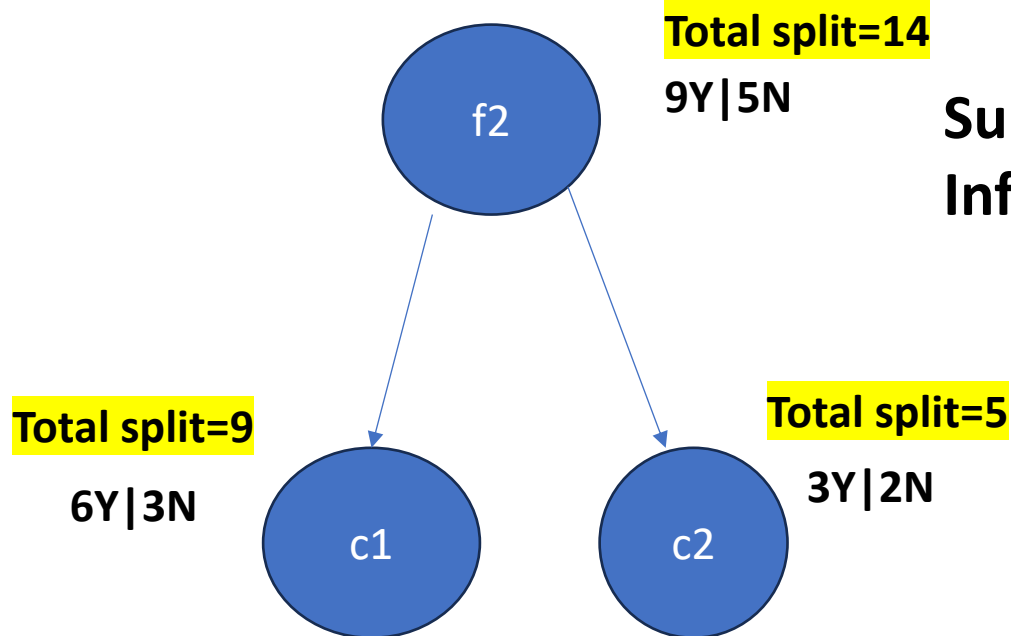


$$\begin{aligned} \text{Information Gain}(S, f1) &= 0.94 - \left[\frac{8}{14} 0.81 + \frac{6}{14} * 1 \right] \\ &= \mathbf{0.049} \end{aligned}$$

Two orange arrows point from the terms $\frac{8}{14} 0.81$ and $\frac{6}{14} * 1$ in the equation above to the $\frac{|S_v|}{|S|}$ and $\text{Entropy}(S_v)$ components of the general formula in the first block, respectively.

Information Gain:

$$\text{Information Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \left(\frac{|S_v|}{|S|} \cdot \text{Entropy}(S_v) \right)$$



Suppose

Information Gain(S,f2) = **0.051**

.....

Information Gain(S,f2) = 0.051 > Information Gain(S,f1) = 0.049

Then we have to start our split from feature f2

Decision tree Split for Numerical features

Data set

Age	Purchased (Target)
22	No
25	No
28	Yes
32	Yes
35	Yes

Sorted Age values: **22, 25, 28, 32, 35**

Possible thresholds (**midpoints**):

23.5, 26.5, 30, 33.5

Steps to Split for Numerical Features:

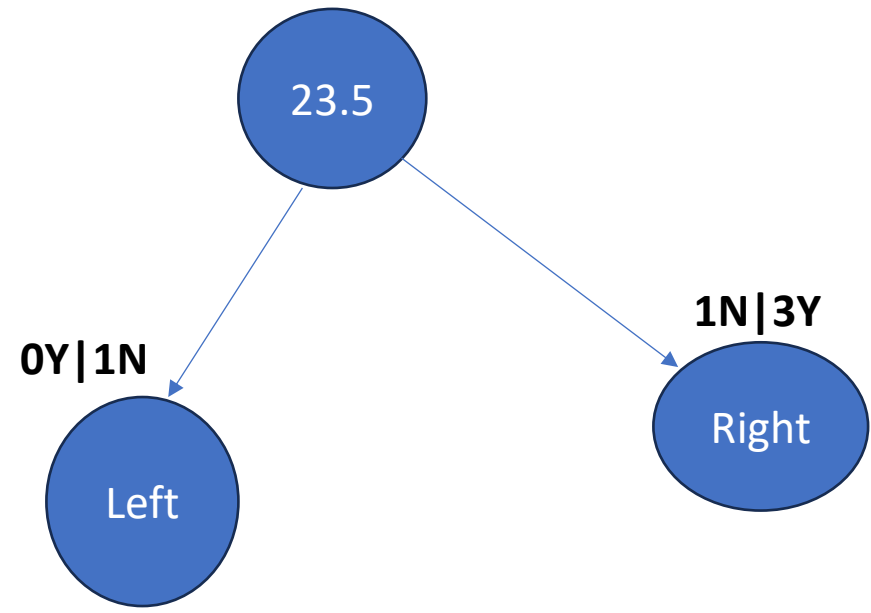
- Identify the feature to split on.
- Sort the data by the selected feature.
- Test each threshold (**midpoints between consecutive values**) as a split point.
- Calculate the impurity for the split using the chosen criterion.
- Choose the split point with the best impurity score (**lowest impurity**).

.....

Threshold = 23.5

Left Subset: [22], Targets: [No]

Right Subset: [25, 28, 32, 35], Targets: [No, Yes, Yes, Yes]



.....

Formula for **Gini Index**:

$$Gini = 1 - \sum (p_i^2)$$

Where p_i is the proportion of a class in a node.

Gini for Left Subset (Targets: [No]):

$$Gini_{left} = 1 - \left(\frac{1}{1}\right)^2 = 0$$

Gini for Right Subset (Targets: [No, Yes, Yes, Yes]):

$$P_{No} = \frac{1}{4} \quad P_{yes} = \frac{3}{4}$$

$$Gini_{right} = 1 - \left(\frac{1}{4}\right)^2 - \left(\frac{3}{4}\right)^2$$

$$= 1 - \left(\frac{1}{16}\right)^2 - \left(\frac{9}{16}\right)^2 = \frac{6}{16} = 0.375$$

.....

Weighted Gini Formula:

$$Gini = \frac{n_{left}}{n_{total}} \cdot Gini_{left} + \frac{n_{right}}{n_{total}} \cdot Gini_{right}$$

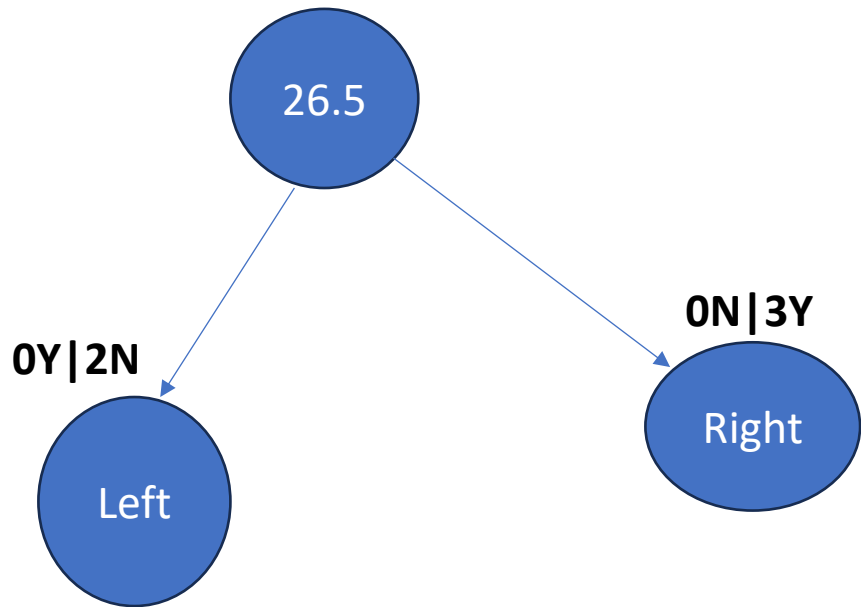
$$Gini = \frac{1}{5} \cdot 0 + \frac{4}{5} \cdot 0.375 = 0.3$$

.....

Threshold = 26.5

Left Subset: [22,25], Targets: [No,No]

Right Subset: [28, 32, 35], Targets: [Yes, Yes, Yes]



.....

Formula for **Gini Index**:

$$Gini = 1 - \sum (p_i^2)$$

Where p_i is the proportion of a class in a node.

Gini for Left Subset (Targets: [No,No]):

$$Gini_{left} = 1 - \left(\frac{2}{2}\right)^2 = 0$$

Gini for Right Subset (Targets: [Yes, Yes, Yes]):

$$\begin{aligned} Gini_{right} &= 1 - \left(\frac{3}{3}\right)^2 \\ &= 1 - 1 = 0 \end{aligned}$$

.....

Weighted Gini Formula:

$$Gini = \frac{n_{left}}{n_{total}} \cdot Gini_{left} + \frac{n_{right}}{n_{total}} \cdot Gini_{right}$$

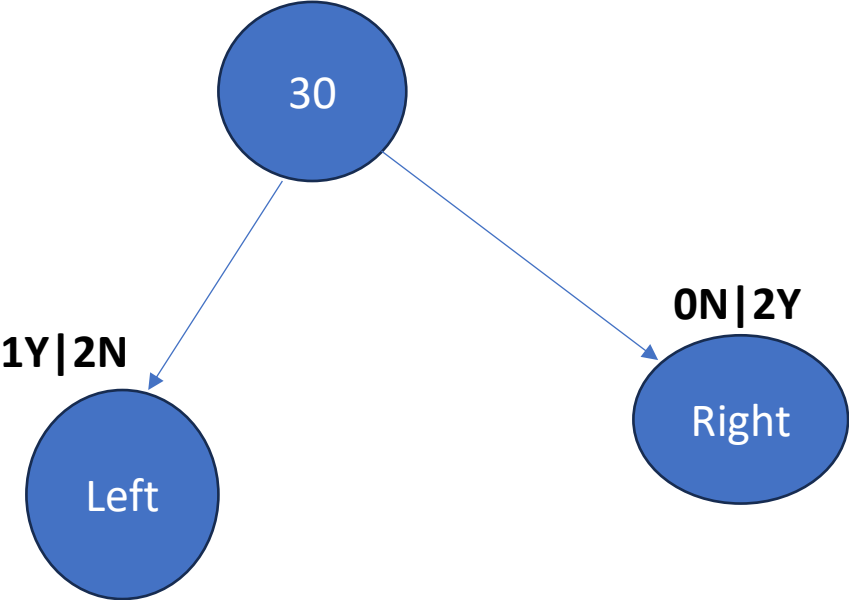
$$Gini = \frac{2}{5} \cdot 0 + \frac{3}{5} \cdot 0 = 0$$

.....

Threshold = 30

Left Subset: [22,25,28], Targets: [**No,No,Yes**]

Right Subset: [32, 35], Targets: [**Yes, Yes**]



.....

Formula for **Gini Index**:

$$Gini = 1 - \sum (p_i^2)$$

Where p_i is the proportion of a class in a node.

Gini for Left Subset (Targets: [No,No,Yes]):

$$\begin{aligned} Gini_{left} &= 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 \\ &= 1 - \left(\frac{4}{9}\right) - \left(\frac{1}{9}\right) = \frac{4}{9} \end{aligned}$$

Gini for Right Subset (Targets: [Yes, Yes]):

$$\begin{aligned} Gini_{right} &= 1 - \left(\frac{2}{2}\right)^2 \\ &= 1 - 1 = 0 \end{aligned}$$

.....

Weighted Gini Formula:

$$Gini = \frac{n_{left}}{n_{total}} \cdot Gini_{left} + \frac{n_{right}}{n_{total}} \cdot Gini_{right}$$

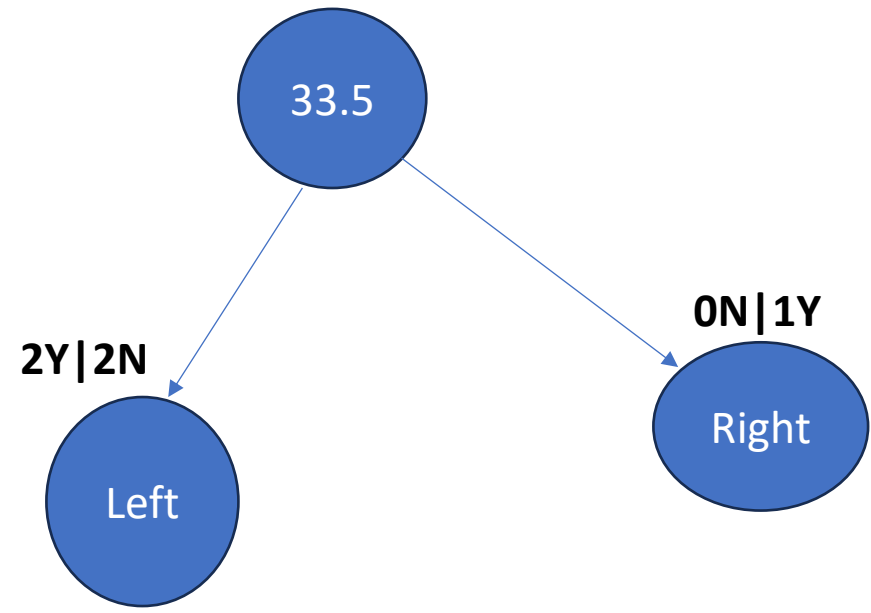
$$Gini = \frac{3}{5} \cdot \frac{4}{9} + \frac{2}{5} \cdot 0 = 0.27$$

.....

Threshold = 33.5

Left Subset: [22,25,28,32], **Targets:** [**No,No,Yes,Yes**]

Right Subset: [35], **Targets:** [**Yes**]



.....

Formula for **Gini Index**:

$$Gini = 1 - \sum (p_i^2)$$

Where p_i is the proportion of a class in a node.

Gini for Left Subset (Targets: [No,No,Yes,Yes]):

$$\begin{aligned} Gini_{left} &= 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2 \\ &= 1 - \left(\frac{4}{16}\right) - \left(\frac{4}{16}\right) = \frac{8}{16} = \mathbf{0.5} \end{aligned}$$

Gini for Right Subset (Targets: [Yes]):

$$\begin{aligned} Gini_{right} &= 1 - \left(\frac{1}{1}\right)^2 \\ &= 1 - 1 = 0 \end{aligned}$$

.....

Weighted Gini Formula:

$$Gini = \frac{n_{left}}{n_{total}} \cdot Gini_{left} + \frac{n_{right}}{n_{total}} \cdot Gini_{right}$$

$$Gini = \frac{4}{5} \cdot 0.5 + \frac{1}{5} \cdot 0 = 0.4$$

Summary of Gini value

Threshold	Gini Index
23.5	0.3
26.5	0.0
30	0.27
33.5	0.4

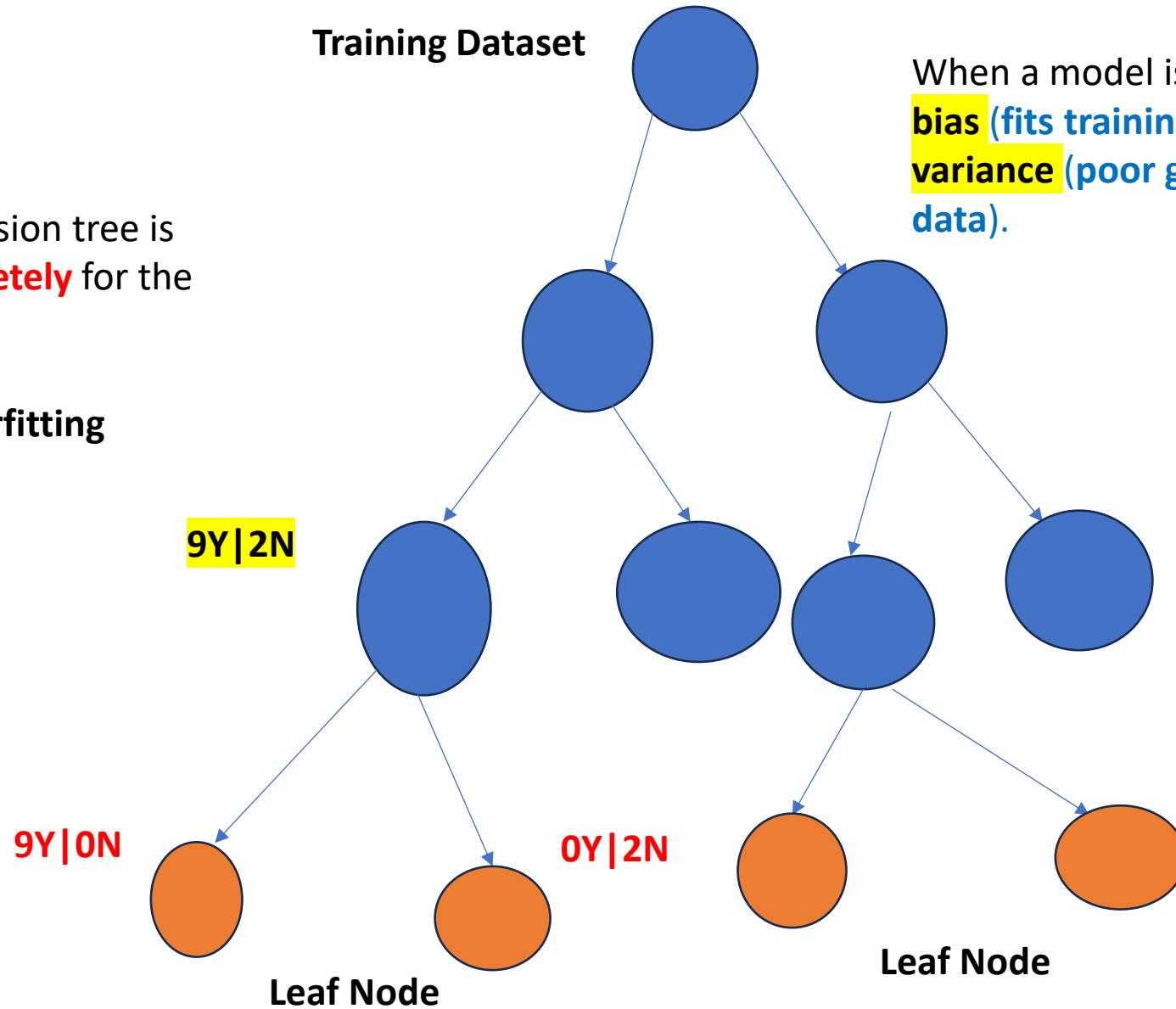
The lowest Gini index is **0.0** at the threshold **26.5**.

Thus, splitting at **Age = 26.5** is the best choice.

Post pruning and Pre Pruning

When the Decision tree is **split** **completely** for the training data.

It leads to **Overfitting**



When a model is **overfitting**, it has **low bias** (fits training data well) but **high variance** (poor generalization to test data).

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In order to **reduce the overfitting** , we are using

- **Post pruning**
- **Pre pruning**

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Post Pruning

1. We first construct the decision tree
2. Prune it with respect to depth

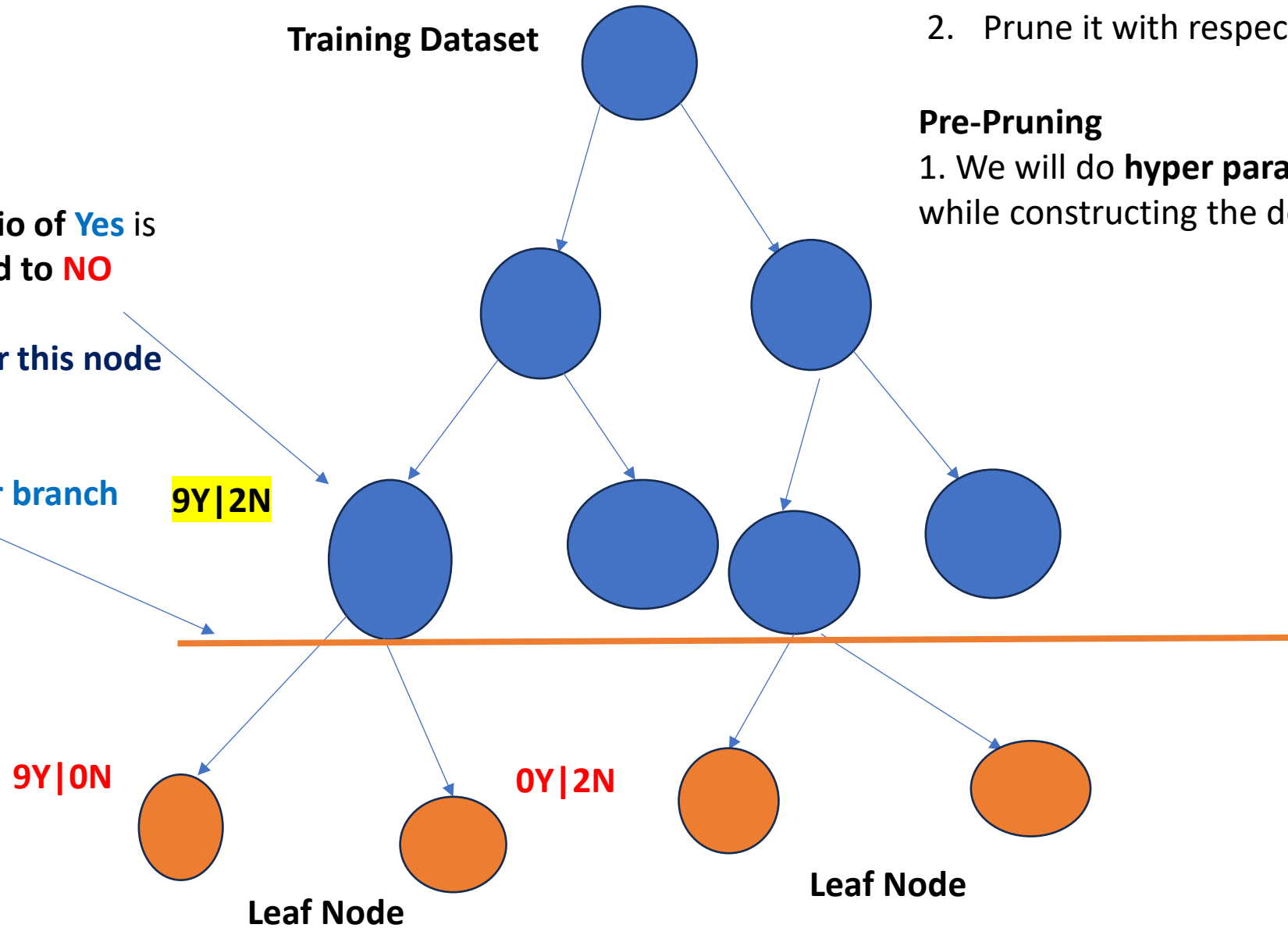
Pre-Pruning

1. We will do **hyper parameter tuning** while constructing the decision tree

In this node the **ratio of Yes** is
very high compared to NO

So, we can consider this node
as leaf node

No need of further branch
We can prune ...



Hyperparameters in a Decision Tree

max_depth:

- The maximum depth of the tree.

min_samples_split:

- The minimum number of samples required to split a node.

min_samples_leaf:

- The minimum number of samples a leaf node must have.

max_features:

- The maximum number of features to consider when looking for the best split.

max_leaf_nodes:

- Limits the maximum number of leaf nodes.

min_impurity_decrease:

- A node will split only if the decrease in impurity (e.g., Gini or entropy) is above this threshold.

Building a Decision Tree using the ID3 (Iterative Dichotomiser 3) algorithm for a weather forecast dataset.

Outlook	Temperature	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rain	Mild	High	False	Yes
Rain	Cool	Normal	False	Yes
Rain	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rain	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rain	Mild	↓ High	True	No

Step-1

The example dataset has the following attributes:

- **Outlook:** Sunny, Overcast, Rain
- **Temperature:** Hot, Mild, Cool
- **Humidity:** High, Normal
- **Windy:** True, False
- **Play:** Yes, No (Target variable)

Step 2: Algorithm

ID3 builds the tree using **Information Gain** based on **Entropy**.

1. Calculate the Entropy of the Target Variable (Play):

$$\text{Entropy} = - \sum p_i \log_2(p_i)$$

Where p_i is the proportion of each class.

- Total instances: 14
- Positive (Play=Yes): 9
- Negative (Play=No): 5

$$\text{Entropy}(Play) = - \left(\frac{9}{14} \log_2 \frac{9}{14} \right) - \left(\frac{5}{14} \log_2 \frac{5}{14} \right) = 0.940$$

.....

2. Compute Entropy and Gain for Each Attribute:

The formula for Information Gain is:

$$\text{Gain}(\textit{Attribute}) = \text{Entropy}(\textit{Play}) - \text{Weighted Entropy of Attribute}$$

Example: Outlook

1. Split by Outlook:

- **Sunny:** 5 instances (2 Yes, 3 No)
- **Overcast:** 4 instances (4 Yes, 0 No)
- **Rain:** 5 instances (3 Yes, 2 No)

.....

2. Calculate Entropy for each split:

- Sunny:

$$\text{Entropy}(\text{Sunny}) = - \left(\frac{2}{5} \log_2 \frac{2}{5} \right) - \left(\frac{3}{5} \log_2 \frac{3}{5} \right) = 0.971$$

- Overcast:

$$\text{Entropy}(\text{Overcast}) = - \left(\frac{4}{4} \log_2 \frac{4}{4} \right) - 0 = 0$$

- Rain:

$$\text{Entropy}(\text{Rain}) = - \left(\frac{3}{5} \log_2 \frac{3}{5} \right) - \left(\frac{2}{5} \log_2 \frac{2}{5} \right) = 0.971$$

3. Weighted Entropy for Outlook:

$$\text{Entropy}(\text{Outlook}) = \left(\frac{5}{14} \cdot 0.971 \right) + \left(\frac{4}{14} \cdot 0 \right) + \left(\frac{5}{14} \cdot 0.971 \right) = 0.693$$

.....

4. Gain for Outlook:

$$\text{Gain}(\textit{Outlook}) = 0.940 - 0.693 = 0.247$$

.....

1. Temperature

Splits:

- **Hot:** 4 instances (2 Yes, 2 No)
- **Mild:** 6 instances (4 Yes, 2 No)
- **Cool:** 4 instances (3 Yes, 1 No)

.....

Entropy for each split:

- Hot:

$$\text{Entropy}(\textit{Hot}) = - \left(\frac{2}{4} \log_2 \frac{2}{4} \right) - \left(\frac{2}{4} \log_2 \frac{2}{4} \right) = 1.0$$

- Mild:

$$\text{Entropy}(\textit{Mild}) = - \left(\frac{4}{6} \log_2 \frac{4}{6} \right) - \left(\frac{2}{6} \log_2 \frac{2}{6} \right) = 0.918$$

- Cool:

$$\text{Entropy}(\textit{Cool}) = - \left(\frac{3}{4} \log_2 \frac{3}{4} \right) - \left(\frac{1}{4} \log_2 \frac{1}{4} \right) = 0.811$$

Weighted Entropy for Temperature:

$$\text{Entropy}(\textit{Temperature}) = \left(\frac{4}{14} \cdot 1.0 \right) + \left(\frac{6}{14} \cdot 0.918 \right) + \left(\frac{4}{14} \cdot 0.811 \right) = 0.911$$

.....

Gain for Temperature:

$$\text{Gain}(\textit{Temperature}) = 0.940 - 0.911 = 0.029$$

.....

2. Humidity

Splits:

- **High:** 7 instances (3 Yes, 4 No)
- **Normal:** 7 instances (6 Yes, 1 No)

Entropy for each split:

- **High:**

$$\text{Entropy}(\text{High}) = - \left(\frac{3}{7} \log_2 \frac{3}{7} \right) - \left(\frac{4}{7} \log_2 \frac{4}{7} \right) = 0.985$$

- **Normal:**

$$\text{Entropy}(\text{Normal}) = - \left(\frac{6}{7} \log_2 \frac{6}{7} \right) - \left(\frac{1}{7} \log_2 \frac{1}{7} \right) = 0.591$$

.....

Weighted Entropy for Humidity:

$$\text{Entropy}(\textit{Humidity}) = \left(\frac{7}{14} \cdot 0.985 \right) + \left(\frac{7}{14} \cdot 0.591 \right) = 0.788$$

Gain for Humidity:

$$\text{Gain}(\textit{Humidity}) = 0.940 - 0.788 = 0.152$$

.....

3. Windy

Splits:

- **True:** 6 instances (3 Yes, 3 No)
- **False:** 8 instances (6 Yes, 2 No)

Entropy for each split:

- **True:**

$$\text{Entropy}(\text{True}) = - \left(\frac{3}{6} \log_2 \frac{3}{6} \right) - \left(\frac{3}{6} \log_2 \frac{3}{6} \right) = 1.0$$

- **False:**

$$\text{Entropy}(\text{False}) = - \left(\frac{6}{8} \log_2 \frac{6}{8} \right) - \left(\frac{2}{8} \log_2 \frac{2}{8} \right) = 0.811$$

.....

Weighted Entropy for Windy:

$$\text{Entropy}(\text{Windy}) = \left(\frac{6}{14} \cdot 1.0 \right) + \left(\frac{8}{14} \cdot 0.811 \right) = 0.892$$

Gain for Windy:

$$\text{Gain}(\text{Windy}) = 0.940 - 0.892 = 0.048$$

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Gains :

- Outlook: 0.247
- Temperature: 0.029
- Humidity: 0.152
- Windy: 0.048

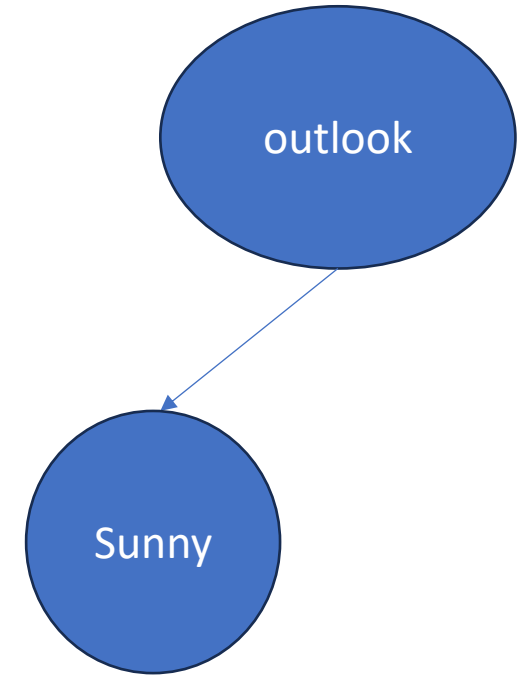
Outlook has the highest gain **(0.247)** and is chosen as the **root node**.

Outlook as the root node.

Step 1: Split by Outlook

1. Subset for Outlook = Sunny

Temperature	Humidity	Windy	Play
Hot	High	False	No
Hot	High	True	No
Mild	High	False	No
Cool	Normal	False	Yes
Mild	Normal	True	Yes



Entropy of this subset:

- Total: 5 instances, Yes = 2, No = 3

$$\text{Entropy}(\text{Sunny}) = - \left(\frac{2}{5} \log_2 \frac{2}{5} \right) - \left(\frac{3}{5} \log_2 \frac{3}{5} \right) = 0.971$$

.....

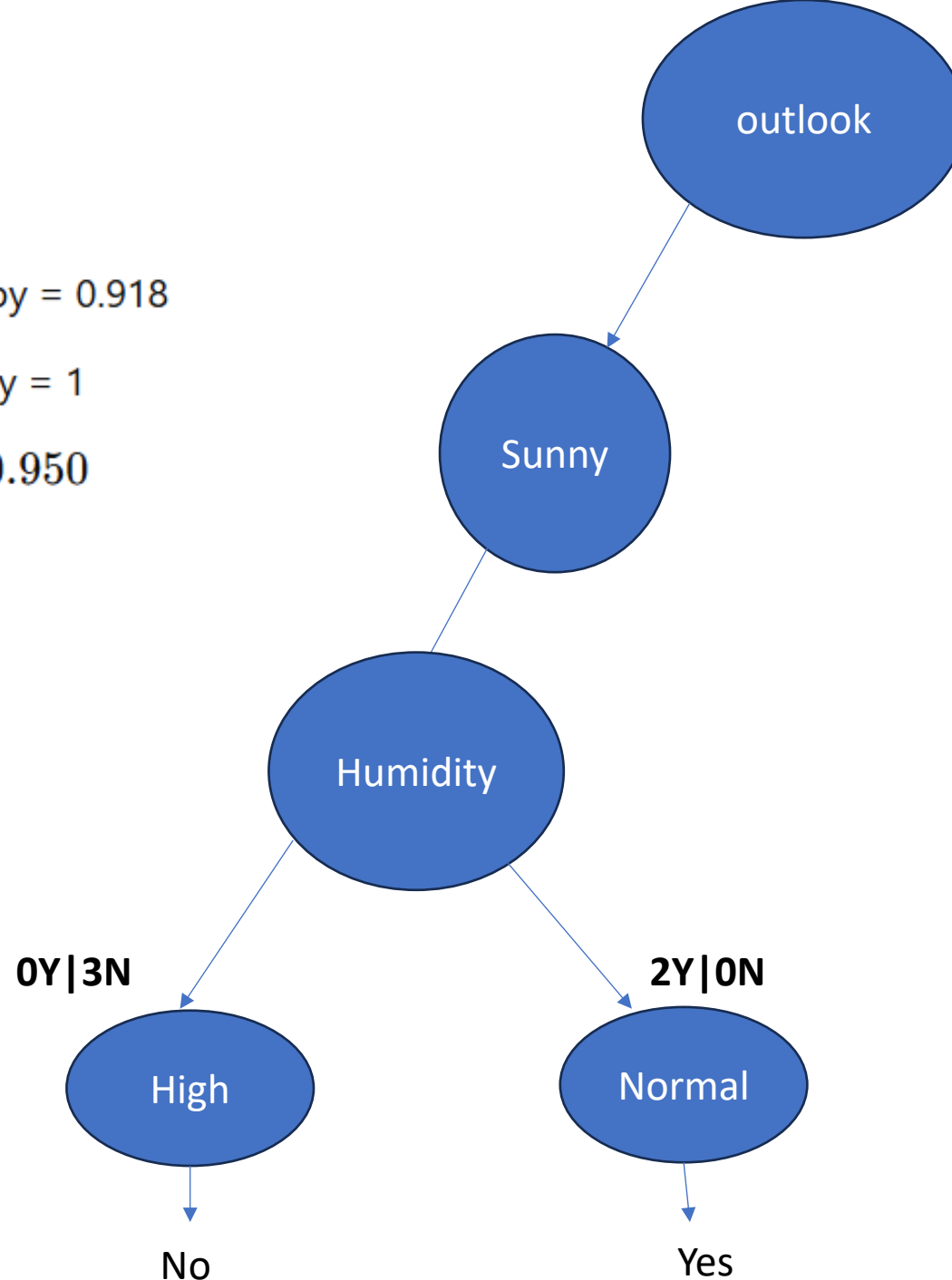
Compute Information Gain for attributes:

- **Humidity:**
 - High (3 instances): Yes = 0, No = 3 $\rightarrow$ Entropy = 0
 - Normal (2 instances): Yes = 2, No = 0 $\rightarrow$ Entropy = 0
 - Weighted Entropy = $\frac{3}{5} \cdot 0 + \frac{2}{5} \cdot 0 = 0$
 - Gain = $0.971 - 0 = 0.971$
- **Temperature:**
 - Hot (2 instances): Yes = 0, No = 2 $\rightarrow$ Entropy = 0
 - Mild (2 instances): Yes = 1, No = 1 $\rightarrow$ Entropy = 1
 - Cool (1 instance): Yes = 1, No = 0 $\rightarrow$ Entropy = 0
 - Weighted Entropy = $\frac{2}{5} \cdot 0 + \frac{2}{5} \cdot 1 + \frac{1}{5} \cdot 0 = 0.4$
 - Gain = $0.971 - 0.4 = 0.571$

.....

- Windy:
 - False (3 instances): Yes = 1, No = 2 → Entropy = 0.918
 - True (2 instances): Yes = 1, No = 1 → Entropy = 1
 - Weighted Entropy = $\frac{3}{5} \cdot 0.918 + \frac{2}{5} \cdot 1 = 0.950$
 - Gain = $0.971 - 0.950 = 0.021$

Best Attribute: **Humidity**

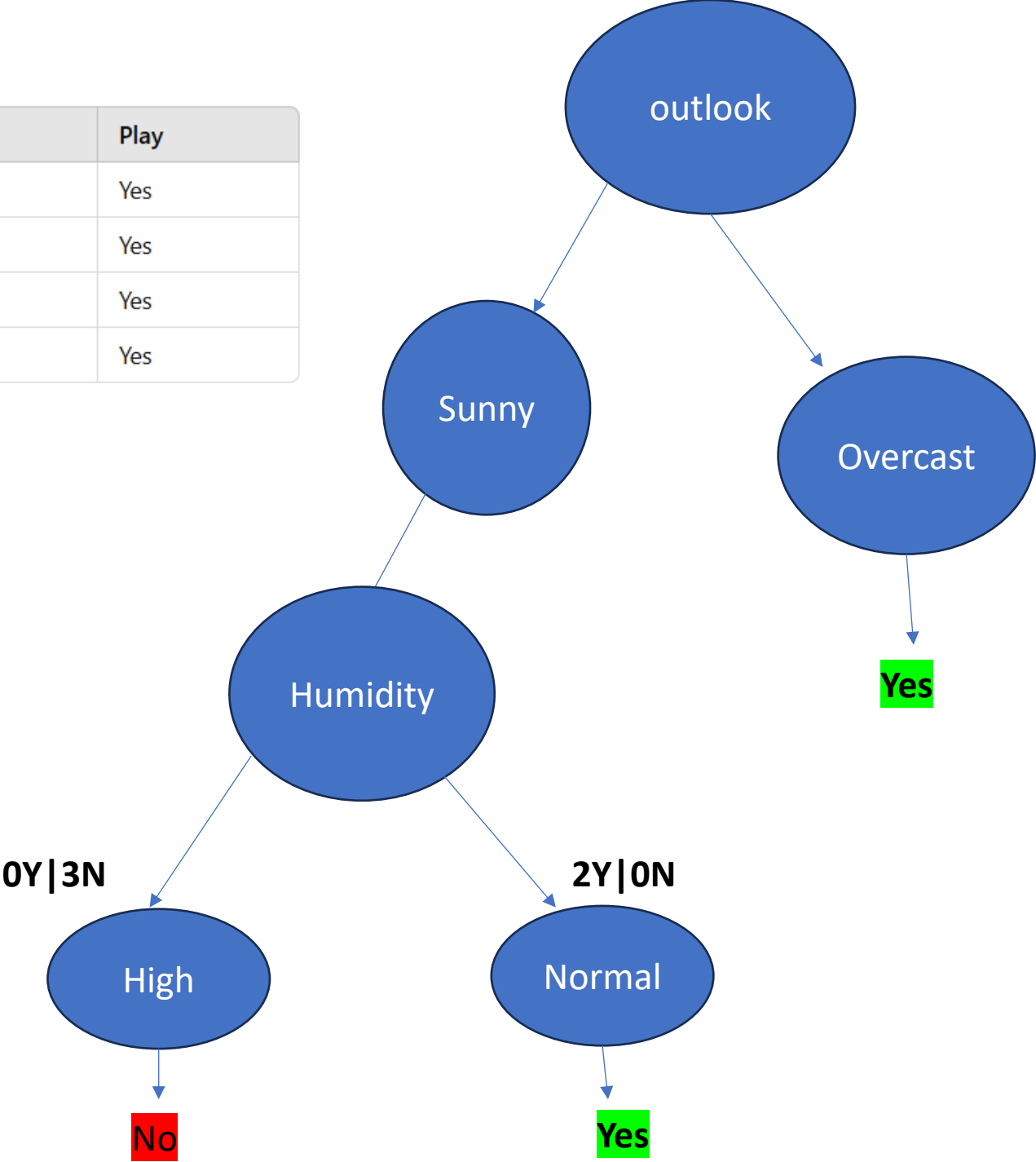


.....

2. Subset for outlook = Overcast

Temperature	Humidity	Windy	Play
Hot	High	False	Yes
Cool	Normal	True	Yes
Mild	High	True	Yes
Hot	Normal	False	Yes

- All instances are **Yes**.
- Entropy = 0 (pure class).
- No further split required.



.....

3. Subset for outlook = Rain

Temperature	Humidity	Windy	Play
Mild	High	False	Yes
Cool	Normal	False	Yes
Cool	Normal	True	No
Mild	Normal	False	Yes
Mild	High	True	No

Entropy of this subset:

- Total: 5 instances, Yes = 3, No = 2

$$\text{Entropy}(\text{Rain}) = - \left(\frac{3}{5} \log_2 \frac{3}{5} \right) - \left(\frac{2}{5} \log_2 \frac{2}{5} \right) = 0.971$$

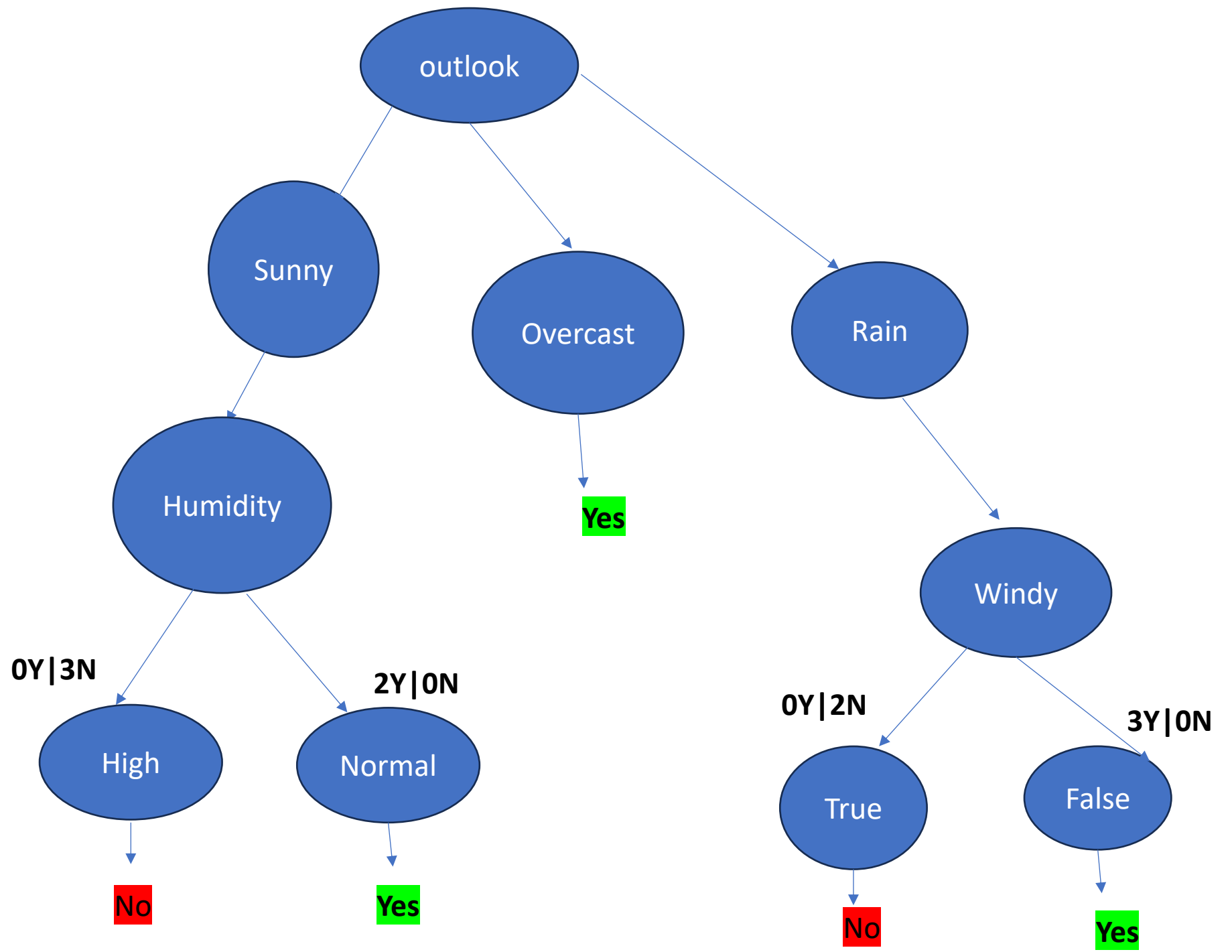
.....

Compute Information Gain for attributes:

- Windy:
 - True (2 instances): Yes = 0, No = 2 $\rightarrow$ Entropy = 0
 - False (3 instances): Yes = 3, No = 0 $\rightarrow$ Entropy = 0
 - Weighted Entropy = $\frac{2}{5} \cdot 0 + \frac{3}{5} \cdot 0 = 0$
 - Gain = $0.971 - 0 = 0.971$
- Humidity:
 - High (2 instances): Yes = 1, No = 1 $\rightarrow$ Entropy = 1
 - Normal (3 instances): Yes = 2, No = 1 $\rightarrow$ Entropy = 0.918
 - Weighted Entropy = $\frac{2}{5} \cdot 1 + \frac{3}{5} \cdot 0.918 = 0.950$
 - Gain = $0.971 - 0.950 = 0.021$

Best Attribute: **Windy**

.....



New Predictions

1. **Sunny, Cool, Normal, False**
2. **Rain, Mild, High, True**

General Rule for Using the Decision Tree:

1. Start at the root node.
2. Follow the branch that matches the attribute value in the instance.
3. Continue until reaching a leaf node (a decision like **Yes** or **No**).
4. The label at the leaf node is the prediction.

.....

1. Sunny, Cool, Normal, False

- Step 1: Start at the root: Outlook = Sunny
 - Follow the branch for Sunny.
- Step 2: At the next level, check Humidity.
 - For this instance, Humidity = Normal, so follow the branch Normal.
- Step 3: The decision for Humidity = Normal under Sunny is Yes.

Prediction: Yes

.....

2. Rain, Mild, High, True

- Step 1: Start at the root: **Outlook** = Rain
 - Follow the branch for **Rain**.
- Step 2: At the next level, check **Windy**.
 - For this instance, **Windy** = **True**, so follow the branch **True**.
- Step 3: The decision for **Windy** = **True** under **Rain** is **No**.

Prediction: No